

Bernard Cooke

SENIOR DATA ENGINEER

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A proactive, results- and customer-oriented systems thinker with close to 7 years' experience in platform, backend and data-centric technology roles in both enterprises and start-ups. I thrive in environments where learning is continuous and autonomy is high. I am motivated by problem-solving at the organisational scale, at the intersection of strategy and operations, planning and execution, technology and people - and equally motivated by creating the space to build others up and support them to do their best work.

Skills

Python development

Extensive Python experience including API development, async/FastAPI, AWS, Apache Airflow, database integration/administration, dbt, polars, PySpark, packaging with uv/Poetry/hatch (including extension modules in e.g. Rust and setting up/administering private registries), profiling, testing/CI, and developer experience (SDK/library development, including full type-annotation semantics)

Software development

Languages including Rust and Golang, leveraging technologies including gRPC, Protocol buffers, HTTP/2 and WebSockets

Platform/Product roadmap development

Stakeholder engagement, workshops, requirements gathering, problem statement definition, work breakdowns/Gantt construction

SRE/Platform support

On-call support, FreshDesk/Jira ServiceDesk workflows, observability with e.g. Grafana, CloudTrail, and definition of KPIs/SLOs/RPO+RTO

AWS Architecture

Experience building AWS Accounts/Organisations from scratch including core networking and route tables, IAM/IAM Identity Centre, DNS, log collection and observability

AI-enabled workflows

for example, Codex and OpenAI

Databases

Experience using PostgreSQL, DuckDB, Oracle, Jena Fuseki, SQLite, MongoDB

CI/CD

Docker/Docker Compose, Jenkins, GoCD, GitHub Actions, JFrog Platform

Cloud/IaC

Terraform, Terragrunt, Ansible, Packer, Chef, Kubernetes/Helm

Other

Git/GitHub, Linux, Bash/shell, CMake, JIRA/Confluence, Agile

Work Experience

Arcturis Data Ltd

SENIOR DATA ENGINEER

Kidlington, UK

Nov. 2023 — Present

Arcturis curates precision real-world data to support the development of new medicines. I joined in the early stages of the establishment of the Engineering department and played a pivotal role in the consolidation of the legacy infrastructure, greenfield build and migration to a replacement platform, and the operation of that platform in a maturing business landscape.

- Developed vision and roadmap for core data pipeline including immediate risk mitigations. Executed on said roadmap, realising significantly better test coverage, reduction in development overhead and defect rate, and ultimately a processing time reduction from days to hours for typical operations.
- Architected and built greenfield AWS Organisation following best practices. Successfully completed whole-company migration from Azure tenant to AWS realising an approx. 50% reduction in cloud expenditure on like-for-like infrastructure alone.
- Developed SOPs and software for company operations in federated data environments (NHS SDEs) and contributed to policy development as the company QMS was established.
- Planned and delivered end-user and stakeholder workshops to shape Engineering roadmap.
- Served as the company's principal point of expertise for Python development.
- Significantly contributed to or wholly authored, deployed and owned multiple internal web applications; many of which were powered by LLM, and one such described in a published whitepaper which I co-authored in Frontiers in Digital Health.

Iotic Labs Ltd (IOTICS)

SOFTWARE ENGINEER II

Remote, UK

Jun. 2022 — Nov. 2023

IOTICS was a startup leveraging the semantic web to solve interoperability challenges and apply FAIR data principles to allow cooperative data ecosystems to form. I was employed as a Software Engineer, developing the Core software that leveraged IoT, blockchain-inspired cryptography, a custom transport layer and cloud- decoupled architecture.

- Implemented and reviewed complex features across core service stack primarily in Python and Golang, with some Rust components, including both REST and gRPC APIs.
- Re-implemented end-to-end CI/CD workflow including code architecture refactoring.
- Rearchitected product deployment, realising an overall 50+% reduction in hosting costs.
- Provided post-sales end-to-end solution design and implementation for customer engagements.
- Delivered on-site technical product training for clients.
- Maintenance of core AWS architecture including load balancers/ASGs, VPC, DNS, ACM, IAM, etc.
- Administered applications such as JIRA/Confluence, GitHub, Okta, JFrog Platform
- Designed and delivered a prototype Azure-hosted product deployment for a prospective enterprise customer demo on a one-week turnaround.

Elsevier Ltd

Kidlington, UK

TECHNOLOGY ASSOCIATE/SOFTWARE ENGINEER II-IV

Sep. 2019 — Jun 2022

Elsevier Ltd is a major global scientific, technical, and medical information provider, part of the RELX Group. Working within the Business Technology Solutions division, I joined as a Technology Associate and was promoted three times to Software Engineer IV during my employment.

- Led development in critical data warehouse migration to Snowflake + AWS, with projected \$2m annual reduction in software/infrastructure costs and over 50% reduction in daily data processing time.
- Designed and implemented of PySpark-based cross-database reconciliation framework and other Python-based data testing utilities.
- Developed and maintained ELT pipelines for multiple financial data warehouses using Oracle/OBIEE, Oracle PL/SQL and Talend.
- Linux server administration of internal Tableau Server, including upgrade and change management and acting as L1-3 support.

Education

University of Manchester

Manchester, UK

MMATH (HONS) IN MATHEMATICS

2015 - 2019

Outside interests

Outside of my day-to-day, I take an interest in several topics, including other specialisms of tech (such as embedded software/robotics, security engineering and malware [both offensive and defensive]), chess and powerlifting.

References available on request.